

Pack a Returns Ready Order SOP

One episode packs, seals, and labels one returns ready order. The carton is already standing open and square on the pack spot with all four top flaps up. It takes two items flat on its floor, a prepaid return label and a return tape strip laid on the items, and an instruction sheet on top; it is held still while the contents are confirmed, has its four top flaps folded in, is sealed with one tape strip along the top seam, carries one shipping label on its label panel, and slides left onto the finished spot.

The order never changes: items, return label, return tape strip, instructions, confirm, close, seal, label, slide out. Nothing goes into the carton under something already in, no flap folds before the contents are confirmed, no strip comes out of the tape dispenser before all four flaps are in, and the shipping label goes on only after the seam strip is pressed flat.

The right gripper takes the two items (in Config R), the return label, the return tape strip, the presented tape strip, and the shipping label; folds the near, far, and right top flaps; presses the seam strip and the shipping label; and supplies the push on the final slide. The left gripper brings the instruction sheet in and (in Config L and M) the two items, folds the left top flap, steadies the carton for every insert, fold, and press the right gripper makes, and guides the slide. Nothing is turned in the air, and a gripper opens only over the spot the thing in it is going to.

The table is set up in one of three ways. Only the item tray moves; the carton, the finished spot, the instruction spot, the return kit spot, the label pad, the tape dispenser, and the tape scrap rest are in the same place in all three.

- **Config L:** the item tray is at the back-left, behind the finished spot.
- **Config M:** the item tray is at the front-center, in front of the pack spot.
- **Config R:** the item tray is at the right edge.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the item tray starts at the back-left (**Config L**), the front-center (**Config M**), or the right edge (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the item tray's side takes each item and lowers it into the carton — the left gripper in Config L and M, the right gripper in Config R — while the other gripper holds the carton. No arm reaches across the table.
- **Fixed roles:** everything else is the same in all three configs — the right gripper fetches the return label, the return tape strip, the seam strip, and the shipping label, folds the near, far,

and right flaps, presses, and pushes the slide; the left gripper brings the instruction sheet, folds the left flap, holds, and guides the slide.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the carton standing on the pack spot at the center, the finished spot and the instruction spot on its left, the return kit spot, label pad, tape dispenser, and tape scrap rest on its right, and the item tray in the start zone for this episode's config.
3. The pack spot is visible from above, so every insert going into the carton, every flap fold, the strip landing on the seam, and the shipping label landing on the label panel can all be seen.
4. Both arms are at home with grippers open.
5. The tabletop is clear of anything but the zones listed below.
6. Both arms reach the pack spot and the finished spot without crossing or reaching a joint limit. The right arm also reaches the return kit spot, label pad, tape dispenser, tape scrap rest, and (Config R) the item tray. The left arm also reaches the instruction spot and (Config L and M) the item tray. Neither arm needs the other's zones. Both grippers can travel beside the carton through the slide without touching it anywhere but its walls.
7. The straight path from the pack spot to the finished spot is bare.

Materials checklist

1. One small **carton** stands square on the **pack spot** with its **near wall** toward the front edge. Its bottom is already taped, it is empty, and all four **top flaps** stand up and clear of the opening. The outer face of its **right flap** carries a printed **label panel**, a plain rectangle large enough for the shipping label with room to spare.
2. One **item tray** holds two **items**, one per **cell**: **cell 1** in the far position and **cell 2** in the near position. Each item is flat sided, light enough for one gripper to lift from above, and small enough that both lie flat side by side on the carton floor with clear space around them. Which two items are used varies between episodes. The count stays at two. The tray stands in the start zone for this episode's config:
 - **Config L:** back-left, behind the finished spot

- **Config M:** front-center, in front of the pack spot, clear of the carton's near wall so the flat hold still fits
 - **Config R:** right edge
3. One **return kit spot** sits at the front right holding two flat pieces side by side: the **return label**, a prepaid printed card lying printed face up on the far side of the spot, and the **return tape strip**, one length of tape still on its **backing card**, backing down and adhesive face up, on the near side of the spot.
 4. One **liner square** is taped flat on the **label pad** at the front right, right of the return kit spot, carrying one **shipping label** printed face up. The label's near edge is already lifted free of the liner so a gripper takes the lifted edge and the label comes away on a straight lift.
 5. One **instruction sheet**, a flat printed sheet, lies on the **instruction spot** at the front left, printed face up, its long edges running left to right. It fits flat inside the carton with room to spare.
 6. One tabletop **tape dispenser** stands to the right of the pack spot. It contains enough tape for the episode and any retries and automatically presents one strip at a time cut to the seam length, adhesive face down, with a stiff **grab tab** accessible to the right gripper. Its stock is not counted during the episode.
 7. One bare **tape scrap rest** sits beside the dispenser for a strip that must be scrapped.
 8. The pack spot, the finished spot, and the path between them are bare, clean, and dry.

Workspace layout

- **Pack spot:** the center of the table, where the carton stands and every insert, fold, seal, and label happens. It holds the carton and nothing else.
- **Finished spot:** the clear unmarked area immediately left of the pack spot. The packed carton stands here at the end of the episode.
- **Instruction spot:** front left, one instruction sheet lying flat and printed face up.
- **Item tray:** the start zone — back-left (**Config L**), front-center (**Config M**), or right edge (**Config R**); two cells, cell 1 far and cell 2 near.
- **Return kit spot:** front right, the return label on the far side and the return tape strip on its backing card on the near side.
- **Label pad:** front right, right of the return kit spot, one liner square taped flat with one shipping label on it, near edge lifted free.
- **Tape dispenser:** right of the pack spot, presenting one cut strip at a time by its grab tab.
- **Tape scrap rest:** beside the dispenser, where a damaged or badly laid strip is placed.

Everything the right gripper fetches is staged on the right. The instruction sheet the left gripper brings is on the left, and in Config L and M the item tray is on the left gripper's side too. The pack spot and the finished spot are shared, and both grippers reach the carton in either position. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the right gripper fetches the two items (Config R), the return label, the return tape strip, the presented tape strip, and the shipping label; folds the near, far, and right top flaps; presses the seam strip and the shipping label; and supplies the push on the slide. The left gripper brings the instruction sheet in and the two items (Config L and M), folds the left top flap, holds the carton for the right gripper's inserts, folds, and presses, and guides the slide. The right gripper takes the hold only while the left gripper lowers something in: the instruction sheet, and in Config L and M the two items. Neither gripper takes over the other's work.
- **Carton:** the small open box standing on its taped bottom. Its **near wall** faces the front edge and its **far wall** faces away.
- **Top flaps:** the four flaps standing up around the carton opening, named for the wall they hinge from: **near flap**, **far flap**, **left flap**, **right flap**.
- **Top seam:** the line down the middle of the closed top where the left flap and the right flap meet. It runs near to far and it is the only place a seam strip goes.
- **Label panel:** the printed rectangle on the outer face of the right flap. It sits clear of the top seam and it is the only place the shipping label goes.
- **Item:** one product going into the carton. Item 1 comes out of cell 1 and item 2 out of cell 2. Both lie flat on the carton floor side by side, and neither ever rides on the other.
- **Start zone:** where the item tray stands at the start of the episode — back-left (**Config L**), front-center (**Config M**), or right edge (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Return label:** the prepaid printed card. It lies flat on the items, printed face up.
- **Return tape strip:** one length of tape on its backing card. It goes in whole, backing down and adhesive face up, and it is carried and placed by its backing card. The tape is never lifted off the backing and no gripper touches its adhesive face.
- **Instruction sheet:** the flat printed sheet. It goes in last and lies flat on the return tape strip, printed face up, so it is the first thing seen when the carton is opened.
- **Contents stack:** what is in the carton and the order it goes in: the two items flat on the carton floor side by side, the return label flat on the items, the return tape strip flat on the return label, and the instruction sheet flat on top. Nothing goes in under something already in.
- **Tape dispenser: Unvalidated at the station.** The tabletop machine automatically advances and cuts one strip ready to use after the presented strip is removed. The grippers do not operate its cutter or reach inside it.
- **Seam strip:** one cut length of tape presented by the dispenser. Its **grab tab** is the stiff end carrying no adhesive, and it is the only part of a seam strip a gripper touches. A seam strip touched anywhere else is scrapped.

- **Hangs straight:** the seam strip dangling free and flat from the lifted grab tab, not curled, twisted, or stuck to itself.
- **Laid along the seam:** the far end of the hanging seam strip set down on the far end of the top seam, then drawn toward the front edge along the seam so the strip settles onto it behind the gripper, with the tab released past the near end.
- **Shipping label:** the one label on the liner square. It is taken by its lifted near edge, comes away from the liner on a straight lift, is carried with the adhesive face down, and is lowered once onto the label panel.
- **Flat hold:** a gripper closed and resting on the carton's near wall to keep it from shifting while the other gripper works. It goes on before the working gripper moves and comes off after the working gripper is clear.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower in:** carrying level over the opening, lowering straight down until the thing is resting on what is already in the carton, and releasing once it is resting. Nothing is dropped in from above the rim.
- **Lower on:** carrying level over the target, lowering straight down until the thing is resting on it, and releasing once it is resting.
- **Left slide: Unvalidated at the station.** Both grippers closed before contact. The right gripper presses the middle of the carton's right wall toward the finished spot while the left gripper travels against the left wall as a guide. The carton keeps its facing and its base stays flat on the table.
- **Release point:** inside the carton, on the tape scrap rest, on the seam strip after it is resting on the seam, or on the shipping label after it is resting on the label panel. A gripper opens nowhere else. The carton is moved with closed grippers and is never held or released.

Steps

Steps 1 to 8 pack the carton, in that order. Step 9 ends the episode once the packed carton stands on the finished spot. Only Step 1 depends on the config: in Config L and M the **left gripper** takes each item from the item tray while the right gripper holds the carton; in Config R the **right gripper** takes each item while the left gripper holds. Every other line is the same in all three configs.

Step 1: Load the two items

Goal: both items lying flat on the carton floor side by side.

Look where the item tray is before reaching for the first item. The holding gripper takes its **flat hold** on the carton's near wall before the first item and keeps it through both.

1.1 Item 1

- **IF the item tray is at the back-left (Config L):** the **right gripper** takes the flat hold. The **left gripper** closes on **item 1** in **cell 1**, lifts it straight up, carries it level forward and right over the carton opening, lowers it in until it is resting flat on the carton floor, and releases.
- **IF the item tray is at the front-center (Config M):** the **right gripper** takes the flat hold. The **left gripper** closes on **item 1** in **cell 1**, lifts it straight up, carries it level back over the carton opening, lowers it in until it is resting flat on the carton floor, and releases.
- **IF the item tray is at the right edge (Config R):** the **left gripper** takes the flat hold. The **right gripper** closes on **item 1** in **cell 1**, lifts it straight up, carries it level left over the carton opening, lowers it in until it is resting flat on the carton floor, and releases.

Then, in all three:

- Item 1 goes to the far half of the carton floor, leaving the near half clear for item 2.

1.2 Item 2

- **IF Config L or M:** the **left gripper** closes on **item 2** in **cell 2**. **IF Config R:** the **right gripper** closes on it. The holding gripper keeps its flat hold.
- Lift it straight up, carry it level over the carton opening, lower it in until it is resting flat on the carton floor beside item 1, and release.
- Item 2 goes to the near half of the carton floor. It never lands on item 1.
- **IF Config L or M:** once the left gripper is clear of the opening, the **right gripper** releases the flat hold and the **left gripper** takes it on the near wall, ready for Step 2.

Take one item per trip. Item 1 is fully in and released before item 2 is picked.

Check: both items lie flat on the carton floor side by side, neither standing on edge, leaning on a wall, nor resting on the other, and the item tray is empty. If an item landed on edge, leaning, or on the other, lift it clear and put it in again.

Step 2: Lay in the return label and the return tape strip

Goal: the return label lying flat and printed face up on the items, and the return tape strip lying flat on the return label with its adhesive face up.

The **left gripper** keeps its **flat hold** on the near wall through both inserts.

2.1 Return label

- The **right gripper** closes on the near edge of the **return label** on the return kit spot, lifts it straight up, carries it level over the carton opening, lowers it in until it is resting flat on the items, and releases.
- It goes in printed face up, the way it lay on its spot. It is not turned, folded, or curled on the way in.

2.2 Return tape strip

- The **right gripper** closes on the near edge of the **backing card** of the **return tape strip**, lifts it straight up, carries it level over the carton opening, lowers it in until it is resting flat on the return label, and releases.
- It goes in whole, backing down and adhesive face up. The tape is never lifted off its backing and the gripper touches nothing but the backing card.

Take one piece per trip. The return label is fully in and released before the return tape strip is picked.

Check: the return label lies flat on the items printed face up, the return tape strip lies flat on the return label with its tape still on its backing and its adhesive face up, no corner of either is folded under or riding a wall, and the return kit spot is empty. If either landed folded, face down, or under something, lift it clear and put it in again.

Step 3: Lay in the instruction sheet

Goal: the instruction sheet lying flat and printed face up on top of the contents stack.

- The **left gripper** releases its flat hold. The **right gripper** takes a **flat hold** on the carton's near wall and keeps it for the whole step.
- The **left gripper** closes on the near edge of the **instruction sheet** on the instruction spot, lifts it straight up, carries it level over the carton opening, lowers it in until it is resting flat on the return tape strip, and releases.
- It goes in printed face up, the way it lay on its spot. It is not turned, folded, or curled on the way in.
- The **right gripper** releases the flat hold once the left gripper is clear of the opening.

Check: the instruction sheet lies flat and printed face up on top of everything else in the carton, with no corner folded under or riding a wall, and the instruction spot is empty. If it landed face down, folded, or under the return tape strip, lift it clear and put it in again.

Step 4: Confirm the contents

Goal: the whole contents stack seen from above with both grippers out of the way, so what went in and what order it went in can be read.

- Lift both grippers clear of the carton opening.
- Keep them clear for 2 seconds with the pack spot in frame, so the printed face of the instruction sheet and the layers under it can be read from above. Nothing is held, nudged, or moved during those 2 seconds.
- If the instruction sheet is face down, folded, or riding a wall, or a layer is out of the contents stack order, correct it now by lifting the wrong piece clear and putting it in again, then keep both grippers clear for 2 seconds again.

Nothing is folded, sealed, or fetched until this confirm is done.

Check: with both grippers clear of the opening, the instruction sheet is flat and printed face up on top, and the return tape strip, the return label, and the two items are visible under it in that order.

Step 5: Close the four top flaps

Goal: all four top flaps folded in flat, with the top seam running near to far down the middle and the label panel face up.

Fold in this order and no other. Every fold is a fold down onto the flaps already in, never a press on the carton contents.

5.1 Near flap, then far flap

- The **left gripper** takes a **flat hold** on the carton's near wall and keeps it for the whole step.
- The **right gripper** pinches the free edge of the **near flap**, folds it away from the front edge down over the opening, and releases.
- The **right gripper** pinches the free edge of the **far flap**, folds it toward the front edge down over the near flap, and releases.

Expected state: the opening covered near to far, the left and right flaps still standing up.

5.2 Left flap, then right flap

- The **left gripper** lifts off the near wall, pinches the free edge of the **left flap**, folds it right down over the flaps already in until it reaches the middle, releases, and goes back to its flat hold on the near wall.
- The **right gripper** pinches the free edge of the **right flap**, folds it left down over the flaps already in until its edge meets the left flap along the middle, and releases.

Check: all four flaps lie flat and down, no flap edge stands proud or is trapped under another flap, the left and right flaps meet along the middle in one straight top seam running near to far, and the label panel on the right flap is face up and clear of the seam. If a flap stands proud or the seam is off the middle, lift that flap and fold it again.

Step 6: Seal the top seam

Goal: one seam strip laid down the top seam and pressed flat, with the carton still square on the pack spot.

- The **left gripper** keeps its **flat hold** on the near wall for the whole seal.
- The **right gripper** pinches the **grab tab** of the seam strip presented by the **tape dispenser**, lifts it straight up until it **hangs straight**, and carries it level over the carton. The dispenser presents the next strip after this one clears it.

- Lay it **along the seam**: set the far end of the strip down on the far end of the top seam, draw the tab toward the front edge along the seam so the strip settles onto it behind the gripper, and release the tab once it lies past the near end.
- The **right gripper** presses along the strip from far to near in one stroke, without coming off the tape, until it lies flat with no lifted end and no bubble standing up.
- Release the right gripper, then the left.

A seam strip is touched only by its grab tab. If it folds onto itself before landing, place it on the tape scrap rest and take the next presented strip. If it lands off the seam or folds on the carton, peel it off, place it on the scrap rest, and take the next presented strip. A scrapped strip is never reused.

Check: one strip runs down the top seam from end to end, lies flat with no lifted end, and the flaps stay down when the carton moves. The label panel is still clear of tape, the dispenser has presented another strip, any scrapped strip is on the tape scrap rest, and no strip is stuck to a gripper.

Step 7: Apply the shipping label

Goal: one shipping label resting flat and printed face up on the label panel, clear of the top seam.

- The **left gripper** takes a **flat hold** on the near wall and keeps it for the whole step.
- The **right gripper** closes on the **lifted near edge** of the **shipping label** on the label pad, lifts it straight up until it comes away from the liner square and hangs free, and carries it level over the carton with its adhesive face down.
- **Lower it on** to the **label panel** until it is resting inside the panel, square to it and clear of the top seam, then release.
- The **right gripper** presses along the label from far to near in one stroke, without coming off the label, until it lies flat with no lifted corner and no bubble standing up.
- Release the right gripper, then the left.

The label is touched only by its lifted edge and it is not turned on the way over. If it comes away curled, folded onto itself, or stuck to the gripper, place it on the tape scrap rest and take a fresh liner square from the label pad. A scrapped label is never reused, and a label that lands on the seam or off the panel is peeled off and scrapped rather than slid into place.

Check: one shipping label lies flat and printed face up inside the label panel, square to it, clear of the top seam and the seam strip, with no lifted corner. The liner square on the label pad is bare, any scrapped label is on the tape scrap rest, and no label is stuck to a gripper.

Step 8: Slide the packed carton onto the finished spot

Goal: the packed carton standing square on the finished spot, sealed seam and shipping label up.

- Close both grippers before either one touches the carton.
- The **right gripper** presses against the middle of the carton's **right wall**. The **left gripper** rests against the middle of its **left wall** as a guide.
- Make one slow **left slide** until the carton stands centered on the **finished spot**. Keep its base flat on the table and its facing unchanged.
- Stop once it stands square, then lift both grippers straight clear.

Do not pinch, lift, tip, or turn the carton, and never move it by a flap or by the shipping label.

Check: the packed carton stands square on the finished spot with its seam and shipping label up and its near wall still toward the front edge, and the pack spot is bare. If it stops short or starts to turn, slide it right back to the pack spot with the push on the left wall and the guide on the right wall, then repeat the left slide.

Step 9: Confirm the packed carton and end the episode

- Confirm the carton stands on the finished spot, closed with all four flaps down, one strip along its top seam, and one shipping label flat inside the label panel; and confirm the item tray, return kit spot, instruction spot, and liner square are empty, the pack spot is bare, and no tape or label is stuck to a gripper.
- Return both arms home, clear of the finished spot and the pack spot, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Peel the shipping label off the label panel and the strip off the top seam, open all four flaps, and set both aside for disposal.
2. Take the instruction sheet, the return tape strip, the return label, and the two items out of the carton.
3. Check the carton for a crushed wall, a torn flap, a bottom tape that has let go, or a label panel torn by the peel. Replace it if it will not stand square with its flaps standing up or its panel is no longer readable.
4. Stand the carton square on the pack spot, empty, bottom taped, near wall toward the front edge, all four top flaps standing up and the label panel on the outer face of the right flap. Leave the path to the finished spot bare.
5. Load the item tray, one item in cell 1 and one in cell 2, each lying flat and clear of the cell walls, and stand it in the start zone for the next episode's config — back-left (Config L),

front-center (Config M), or right edge (Config R). Replace any item that will not lie flat on the carton floor.

6. Lay a fresh return label on the far side of the return kit spot, printed face up, and a fresh return tape strip on its backing card on the near side, backing down and adhesive face up. Replace either if it is creased, torn, or curled, or if the tape has started to lift off its backing.
7. Lay a fresh instruction sheet flat on the instruction spot, printed face up. Replace it if it is creased or torn.
8. Tape a fresh liner square flat on the label pad carrying one shipping label printed face up, and lift the label's near edge free of the liner. Check that the label comes away on a straight lift without curling.
9. Remove all tape scraps and scrapped labels from the scrap rest. Check that the tape dispenser contains enough tape for one episode and retries, cuts each strip to the full seam length, and presents the next strip by a clean grab tab.
10. Wipe the pack spot, the finished spot, and the path between them, and confirm they are clean and dry. A dirty base or tabletop can make the carton catch or turn during the slide.
11. Vary which two items are used between episodes rather than running the same pairing every time. The count stays at one carton, two items, one return label, one return tape strip, one instruction sheet, and one shipping label; seam tape comes from the dispenser on demand.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R (the item tray at the right edge, emptied by the right gripper). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the item tray is not in the start zone for the config; a gripper reaches across the table for an item; the gripper on the tray's side takes the flat hold instead of the item; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper takes each item in Config L and M and the right gripper in Config R, while the other gripper holds the carton; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the item tray is before the first reach, then follow that config's IF lines through Step 1.

Violation: Wrong order

- **Visible cue:** the return label goes in before both items, the return tape strip before the return label, the instruction sheet before the return tape strip, a flap folds before the confirm, a seam strip comes out of the dispenser before all four flaps are in, the shipping label goes on before the seam strip is pressed, or the carton moves off the pack spot before it is sealed and labeled.
- **SOP rule broken:** Steps 1 to 8, the order never changes: items, return label, return tape strip, instructions, confirm, close, seal, label, slide out.
- **Coaching note:** items, return label, return tape strip, instructions, confirm, close, seal, label, slide out. Finish the step you are in before starting the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes an item, the return label, the return tape strip, a seam strip, or the shipping label, folds the near, far, or right flap, or makes the seam or label press; or the right gripper brings the instruction sheet in or folds the left flap; or the push and guide roles are swapped during the slide.
- **SOP rule broken:** Steps 1 to 8, the right gripper fetches, performs the primary work, and supplies the push, while the left gripper brings the instruction sheet, folds the left flap, holds, and guides the slide.
- **Coaching note:** right fetches, works, and pushes; left brings the sheet, folds left, holds, and guides.

Violation: Item put in wrong

- **Visible cue:** an item goes in standing on edge, leaning on a wall, or resting on the other item, or both items are put in one on top of the other instead of side by side, and the next insert goes in anyway.
- **SOP rule broken:** Step 1, each item is lowered in until it rests flat on the carton floor, item 1 to the far half and item 2 to the near half, side by side and never one on the other.

- **Coaching note:** both flat on the floor, side by side. Everything above them is scored on this.

Violation: Return label put in wrong

- **Visible cue:** the return label goes in printed face down, folded, curled, standing on edge, riding up a carton wall, or under an item, and the next insert goes in anyway.
- **SOP rule broken:** Step 2.1, the return label is lowered in printed face up and rests flat on the items, and a bad landing is corrected by lifting it clear and putting it in again.
- **Coaching note:** face up, flat on the items, nothing under it but the items.

Violation: Return tape strip handled or put in wrong

- **Visible cue:** the tape is lifted off its backing card, a gripper pinches the tape itself or touches its adhesive face, the strip goes in adhesive face down, folded, or under the return label, or it comes up stuck to a gripper or a carton wall.
- **SOP rule broken:** Step 2.2, the return tape strip goes in whole on its backing card, backing down and adhesive face up, resting flat on the return label, and it is carried and placed by its backing card only.
- **Coaching note:** hold the backing, never the tape. It goes in as one piece, sticky side up.

Violation: Instruction sheet put in wrong

- **Visible cue:** the instruction sheet goes in printed face down, folded, curled, riding up a carton wall, or under the return tape strip or the return label, and the flaps are folded over it anyway.
- **SOP rule broken:** Step 3, the instruction sheet is lowered in printed face up and rests flat on the return tape strip, on top of everything else in the carton.
- **Coaching note:** face up, last thing in, on top of everything. It is the first thing the customer sees.

Violation: Dropped in from above the rim

- **Visible cue:** a gripper opens above the carton opening and an item, the return label, the return tape strip, or the instruction sheet falls or flutters in, instead of being lowered until it is resting.
- **SOP rule broken:** Steps 1, 2, and 3, lower each insert straight down until it is resting on what is already in the carton, then release.
- **Coaching note:** take it all the way down, then let go. Nothing gets dropped in.

Violation: Carton not steadied

- **Visible cue:** neither gripper is on the near wall while an insert goes in, while a flap is folded, or during the seam or label press, or the carton slides, rocks, or shifts under the working gripper.

- **SOP rule broken:** Steps 1, 2, 3, 5, 6, and 7, one gripper takes a flat hold on the near wall before the working gripper moves and keeps it until the working gripper is clear.
- **Coaching note:** the hold goes on first and comes off last. A loose carton ruins the seam.

Violation: Confirm skipped or worked through

- **Visible cue:** a flap is folded straight after the instruction sheet lands with no still scene between them, a gripper stays in or over the opening during the confirm, or something is nudged, pushed down, or rearranged while the confirm is meant to be running.
- **SOP rule broken:** Step 4, lift both grippers clear and keep them clear for 2 seconds with the pack spot in frame before anything is folded.
- **Coaching note:** hands off and count two. The reviewer has to see what went in and in what order.

Violation: Bad contents left uncorrected

- **Visible cue:** the confirm runs with the instruction sheet face down, folded, or riding a wall, or with the stack out of order, and the flaps are folded anyway.
- **SOP rule broken:** Step 4, correct a wrong insert by lifting it clear and putting it in again, then keep both grippers clear for 2 seconds again.
- **Coaching note:** fix it and confirm again. A confirm that confirms the wrong thing is worse than none.

Violation: Flaps folded in the wrong order

- **Visible cue:** a side flap goes in before the near and far flaps, the far flap goes before the near flap, a flap is trapped under one folded later, or a flap is left standing up when the seam strip comes out.
- **SOP rule broken:** Step 5, fold near, then far, then left, then right, each one down over the flaps already in.
- **Coaching note:** near, far, left, right. Every time, no exceptions.

Violation: Top seam off the middle or flaps proud

- **Visible cue:** the left and right flaps overlap heavily, leave a gap, or meet well off the middle, or a flap edge stands proud of the top, and the seam strip goes on anyway.
- **SOP rule broken:** Step 5.2, the left and right flaps fold in until they meet along the middle in one straight top seam running near to far, and a proud or crooked flap is lifted and folded again.
- **Coaching note:** get the seam straight down the middle before you reach for the tape.

Violation: Seam strip handled off the grab tab

- **Visible cue:** a gripper pinches the seam strip anywhere but its grab tab, touches the adhesive face, or reaches inside the tape dispenser instead of taking the presented tab.

- **SOP rule broken:** Step 6, the grab tab is the only part of a seam strip a gripper touches.
- **Coaching note:** take only the tab presented by the dispenser.

Violation: Seam strip laid off the seam or reused after a bad landing

- **Visible cue:** the seam strip lands at an angle, off to one side, short of either end, across the label panel, or folded onto itself, and is reused instead of being placed on the tape scrap rest.
- **SOP rule broken:** Step 6, lay the seam strip along the whole seam; scrap a damaged or badly laid strip and take the next strip presented by the dispenser.
- **Coaching note:** tape sticks once. Move the bad strip to the scrap rest and take the next one.

Violation: Seam not pressed

- **Visible cue:** the seam strip is laid but never pressed, pressed only at one end, pressed in several separate dabs, or left with an end lifting or a bubble standing up when the carton moves.
- **SOP rule broken:** Step 6, press along the strip from far to near in one stroke without coming off the tape until it lies flat with no lifted end.
- **Coaching note:** one stroke, all the way across, without leaving the tape.

Violation: Shipping label handled wrong

- **Visible cue:** a gripper pinches the shipping label anywhere but its lifted near edge, touches its adhesive face, drags it off the liner sideways instead of lifting it straight up, or turns or flips it between the lift and the placement.
- **SOP rule broken:** Step 7, the shipping label is taken by its lifted near edge, comes away on a straight lift, and is carried level with its adhesive face down and its facing unchanged.
- **Coaching note:** lifted edge only, straight up, and it lands the way it left the pad.

Violation: Shipping label off the panel or on the seam

- **Visible cue:** the label lands outside the label panel, hangs over a carton edge, sits crooked to the panel, covers part of the top seam or the seam strip, or is slid into place after landing instead of being peeled off and scrapped.
- **SOP rule broken:** Step 7, the label is lowered on until it rests inside the label panel, square to it and clear of the top seam, and a label that lands on the seam or off the panel is peeled off and scrapped.
- **Coaching note:** inside the panel, square, off the seam. Once it lands it does not slide.

Violation: Shipping label not pressed

- **Visible cue:** the label is laid but never pressed, pressed only at one end, pressed in several separate dabs, or left with a lifting corner or a bubble standing up when the carton moves.

- **SOP rule broken:** Step 7, press along the label from far to near in one stroke without coming off the label until it lies flat with no lifted corner.
- **Coaching note:** one stroke across the whole label. A lifting corner will not scan.

Violation: Carton slid wrong

- **Visible cue:** the carton is pinched, lifted, tipped, turned, moved by a flap or by the shipping label, or pushed without the other gripper guiding its side wall.
- **SOP rule broken:** Step 8, both grippers are closed before contact, the right gripper supplies the push, the left gripper guides, and the carton base stays flat on the table.
- **Coaching note:** let the table support the carton; right pushes and left guides.

Violation: Carton left off the finished spot

- **Visible cue:** the packed carton stops short of the finished spot, overshoots it, stands turned, or is left standing on the pack spot at the end of the slide.
- **SOP rule broken:** Step 8, slide the carton left until it stands square and centered on the finished spot with its facing unchanged.
- **Coaching note:** take it all the way onto the spot and square, in one slow slide.

Violation: More than one thing handled at once

- **Visible cue:** two items are carried in one trip, the return label and the return tape strip are carried together, a second seam strip is taken while one is still held, or the carton is moved while an insert, a strip, or the label is still held.
- **SOP rule broken:** Steps 1, 2, 6, 7, and 8, take one thing per trip and set it down before picking the next.
- **Coaching note:** one piece at a time, every trip.

Violation: Turned in the air

- **Visible cue:** an item, the return label, the return tape strip, the instruction sheet, a seam strip, or the shipping label is rotated or flipped between lift and placement, or the carton turns during the guided slide.
- **SOP rule broken:** Steps 1 to 8, held items keep their staged orientation, and the carton keeps its facing while sliding on the table.
- **Coaching note:** preserve orientation during picks and use the guide gripper during the carton slide.

Violation: Held item dragged instead of lifted

- **Visible cue:** an item, the return label, the return tape strip, the instruction sheet, a seam strip, or the shipping label drags along its spot during a pick instead of lifting clear. The prescribed carton slide, flap folds, tape draw, and presses are not violations.

- **SOP rule broken:** Steps 1, 2, 3, 6, and 7, lift a held item clear before carrying it sideways.
- **Coaching note:** lift held items; slide only the carton as prescribed.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on an insert, a seam strip, or the shipping label without setting it down first. Travelling contact against a carton wall during the slide is not a grasp.
- **SOP rule broken:** Steps 1, 2, 3, 6, and 7, set a held item down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the item tray, the return kit spot, the instruction spot, the label pad, the tape dispenser, the finished spot, or open table.
- **SOP rule broken:** Steps 1 to 8, a gripper opens inside the carton, on the tape scrap rest, on the seam strip once it rests on the seam, or on the shipping label once it rests on the label panel. The carton is moved with closed grippers.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** an insert is nudged after release, the sealed top is pushed to square a flap, the shipping label is slid after it lands, or the carton is corrected with small pats instead of being returned to the start of its slide.
- **SOP rule broken:** Steps 1 to 8, lift and replace a bad insert, fold a bad flap again, scrap a badly landed strip or label, and return a misplaced carton to the slide's starting point before repeating the full slide.
- **Coaching note:** restart the prescribed motion instead of making small corrective shoves.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** an insert, a seam strip, or the shipping label falls short of its destination; the carton tips; the item tray, return kit spot, label pad, dispenser, or scrap rest shifts; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 8, keep each held item on a clear travel path, the carton on its base, and both arms clear of each other.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the carton open, unsealed, unlabeled, or still on the pack spot, an insert left on its spot, tape or a label stuck to a gripper, an arm away from home, or an arm still over the finished spot.

- **SOP rule broken:** Step 9, confirm the sealed and labeled carton on the finished spot and the empty item tray, return kit spot, instruction spot, liner square, and pack spot; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective carton:** a wall that crushes under a correct push, a base that catches or tips on a clean table, a flap that tears at its fold, bottom tape that lets go, or a label panel printed too small or too faint to place a label against. Replace it before the next episode.
- **Defective item:** one that will not lie flat on a flat carton floor, or that comes apart in the gripper. Replace it before the next episode.
- **Defective return label or instruction sheet:** creased, torn, or curled so it will not lie flat. Replace it before the next episode.
- **Defective return tape strip:** its tape lifts off the backing card under a correct pick, or the backing card is torn. Replace it before the next episode.
- **Label pad fault:** the shipping label will not come away from the liner on a straight lift, comes away curled or torn, or its near edge is not lifted free. Replace the liner square before the next episode.
- **Tape dispenser fault:** it fails to present a strip, cuts a strip shorter than the seam, presents a tab with adhesive on it or out of the gripper's reach, or does not advance after a correct pick. Service it before the next episode.
- **Pack spot, finished spot, item tray, return kit spot, label pad, dispenser, or tape scrap rest shifts out of its zone** during the episode.
- **Carton slides** under a correct flat hold because the pack spot is dusty or slick. Wipe it before the next episode.

Annotation subtasks (from SOP)

1. Lower the two items into the carton flat and side by side
2. Lay the return label on the items face up
3. Lay the return tape strip on the return label
4. Lower the instruction sheet in on top
5. Hold both grippers clear to confirm the contents

6. Fold the near and far top flaps in
7. Fold the left and right top flaps in
8. Lay one tape strip along the top seam
9. Press the seam strip flat
10. Place the shipping label on the label panel
11. Press the shipping label flat
12. Slide the packed carton onto the finished spot
13. Return both arms home and end the episode